



Well Informed

健康知识专栏

难度评级：★★★★☆

How to combine exercise with video games

电子游戏如何与健康运动如何结合？

【 Para.1 】

Exercising can be **tedious**. Video games are fun. What if you could combine them? Playing games while wearing a **virtual-reality** (VR) headset, with a motion-**sensitive** controller in each hand, can be **taxing**. “Pistol Whip” makes you step from side to side and **squat** frequently as you **gun down foes** and **dodge** bullets. “The Thrill of the Fight” is a virtual boxing game that works up a real **sweat**. In “Supernatural” and “Beat Saber” you punch and slice moving targets in time to pumping music. But does any of this **count** as proper exercise?

运动可能会很乏味。电子游戏则充满乐趣。如果能将两者结合起来呢？戴着虚拟现实（VR）头显，双手各握一个动作感应控制器玩游戏，可能会让人筋疲力尽。在游戏《Pistol Whip》中，你需要一边射杀敌人、

文章导读：

电子游戏和运动相结合，可以算得上运动吗？有研究发现运动类虚拟现实（VR）游戏的代谢量不小。另有研究则发现部分运动能够拉高心率。实际减肥效果也未打折扣。但不同运动类游戏的效果存在差异。有研究团队设置了对照组，发现VR类游戏的消耗量更大，但感觉要比实际消耗轻松。但关于此类游戏对于坚持运动有无积极作用，尚未可知。

tedious a. /'ti:diəs/ lasting or taking too long and not interesting 冗长的；啰唆的；单调乏味的；令人厌烦的【考研大纲词汇】

SYN boring

The journey soon became **tedious**.

那次旅行不久就变得乏味起来

virtual-reality n. Virtual reality is an environment that is produced by a computer and seems very like reality to the person experiencing it. 虚拟现实

One day **virtual reality** will revolutionize the entertainment industry.

有朝一日虚拟现实将给娱乐业带来一场革命。

sensitive a. Something that is sensitive to a physical force, substance, or treatment is easily affected by it and often harmed by it. 敏感的【考研大纲词汇】

...a chemical which is **sensitive** to light.

...一种对光敏感的化学品。

taxing a. A taxing task or problem is one that requires a lot of mental or physical effort. 费劲的【考研大纲词汇】

It's unlikely that you'll be asked to do anything too **taxing**.

不大可能叫你去做任何太费劲的事。

squat v. /skwɒt/ If you squat, you lower yourself toward the ground, balancing on your feet with your legs bent. 蹲下

We **squatted** beside the pool and watched the diver sink slowly down.

我们蹲在池边，看潜水员慢慢沉下水。

gun sb down v. [usually passive] to shoot sb, especially killing or seriously injuring them 枪杀，开枪伤害（某人）

foe n. (old-fashioned) (formal) an enemy 敌人；仇敌

dodge v. /dɒdʒ/ to move quickly and suddenly to one side in order to avoid sb/sth 闪开；躲开；避开

He ran across the road, **dodging** the traffic.

他躲开来往的车辆跑过马路。

躲避子弹，一边频繁地左右跨步和下蹲。另一款虚拟拳击游戏《The Thrill of the Fight》则能让人真正大汗淋漓。而在《Supernatural》和《Beat Saber》中，你则需要随着动感十足的音乐节奏，击打并劈砍移动中的目标。但这些真的算得上正规的锻炼吗？

【 Para.2 】

The answer is yes, though with **caveats**. One way to evaluate VR exercise is a measure called **metabolic equivalent** of task (MET). Someone **at rest** expends about one MET. Walking is three METs. Playing football, ten. Anything over six METs is considered “**vigorous**” exercise. Researchers at San Francisco State University determined that playing “The Thrill of the Fight” (which is, in fact, **exhausting**) was a 9.3 MET activity. Other work, published in JMIR Serious Games, found the “Flow” and “Boxing” modes in “Supernatural” also counted as vigorous exercise, at around eight METs.

答案是肯定的，但也附带一些条件。评估VR运动的方法之一是使用“代谢当量”（MET）这一指标。静息状态下，人体消耗约1 MET；步行为3 MET；踢足球则为10 MET。任何超过6 MET的活动都被视为“剧烈”运动。旧金山州立大学的研究人员确定，玩《The Thrill of the Fight》（这款游戏实际上确实非常耗费体力）的代谢当量达到了9.3。发表在《JMIR Serious Games》上的另一项研究发现，《超自然》中的“Flow”和“Boxing”模式也属于剧烈运动，约为8 MET。

【 Para.3 】

In yet another study, published in Games for Health Journal

sweat n. the state of being covered with sweat 出汗；流汗；一身汗【考研英语大纲词汇】

She completed the routine without even working up a **sweat**. 她完成了一套常规动作，连汗都没出。

count v. If something counts or is counted as a particular thing, it is regarded as being that thing, especially in particular circumstances or under particular rules. 看作

No one agrees on what **counts** as a desert.

没有人对于什么可以看作是荒漠有共识。

caveat n. /'keɪv.ət/ A caveat is a warning of a specific limitation of something such as information or an agreement. 警告；提醒

There was one **caveat**: he was not to enter into a merger or otherwise weaken the Roche family's control of the firm.

提醒一下：他不是要参与并购或其他削弱罗技家族对该公司控制的行动。

metabolic a. /'metə'bolɪk/ Metabolic means relating to a person's or animal's metabolism. 新陈代谢的

People who have inherited a low **metabolic** rate will gain weight. 因遗传而新陈代谢速度慢的人会发胖。

equivalent n. /'kwɪvələnt/ If one amount or value is the equivalent of another, they are the same. 等量物；等价物【考研英语大纲词汇】

Mr. Li's pay is the **equivalent** of about \$80 a month.

李先生的报酬大约等于每月80美元。

at rest phrase not moving 静止；不动【考研英语大纲词汇】

At rest the insect looks like a dead leaf.

这种昆虫不动时看上去像一片枯叶。

vigorous a. /'vɪɡərəs/ very active, determined or full of energy 充满活力的；果断的；精力充沛的【考研英语大纲词汇】

SYN energetic

Take **vigorous** exercise for several hours a week.

每周做几个小时剧烈运动。

exhausting a. /ɪg'zɔːstɪŋ/ making you feel very tired 使人疲惫不堪的；令人筋疲力尽的【考研英语大纲词汇】

an **exhausting** day at work 工作得筋疲力尽的一天

treadmill n. [C] an exercise machine that has a moving surface that you can walk or run on while remaining in the same place (锻炼身体的) 跑步机，走步机

aerobic a. /ə'reʊbɪk/ (of physical exercise 健身活动) especially designed to improve the function of the heart and lungs 有氧的；增强心肺功能的

OPP anaerobic

cardio n. cardiovascular exercise 心肺运动

get round phrase = get around To get around a problem or difficulty means to overcome it. 解决【考研英语大纲词汇】

None of these countries has found a way yet to **get around** the problem of the polarization of wealth.

这些国家均未找到一个方案来解决贫富两极分化的问题。

in 2024, participants played a sedentary video game, two VR games, and walked on a **treadmill**, each for ten minutes, while having their oxygen consumption measured. Playing “The Thrill of the Fight” was comparable in **aerobic** intensity to walking on a treadmill, and raised heart rates by a similar amount.

在2024年发表于《健康游戏期刊》（Games for Health Journal）的另一项研究中，参与者分别体验了一款静态电子游戏、两款VR游戏，以及在跑步机上行走，每项活动持续10分钟，同时测量其耗氧量。研究发现，玩《The Thrill of the Fight》的有氧运动强度与在跑步机上行走相当，且心率上升幅度也相似。

【 Para.4 】

So much for **cardio**. What about weight loss? Work published last year in Nature Medicine divided 227 adolescents into five groups: real table tennis, real football, virtual table tennis, virtual football and a control group. There was no significant difference in weight loss between the real and virtual groups. Members of both lost about 5kg after eight weeks. Those in the control group lost none.

心肺效果暂且聊到这。那么减肥效果如何呢？去年发表在《自然·医学》上的一项研究将227名青少年分为五组：真实乒乓球组、真实足球组、虚拟乒乓球组、虚拟足球组和对照组。真实运动组与虚拟运动组在减重效果上没有显著差异。两组成员在八周后均减重约5公斤，而对照组则未减重。

【 Para.5 】

But not all VR games are created equal. The variation in

augment v. /ɔːg'ment/ to augment something means to make it larger, stronger, or more effective by adding something to it. 增加【考研大纲词汇】

While searching for a way to **augment** the family income, she began making dolls.

为求增加家庭收入，她开始做起了玩具娃娃。

stronghold n. If you say that a place or region is a stronghold of a particular attitude or belief, you mean that most people there share this attitude or belief. (大多数人具有共同态度或信仰的) 大本营

Florida is a **stronghold** for pro-choice activists.

佛罗里达是一个主张堕胎合法的活跃分子们的“大本营”。

deadlift n. a lift made from a standing position 硬举

add-on n. a thing that is added to sth else 附加物；附加装置

The company offers scuba-diving as an **add-on** to the basic holiday price.

这家公司提供带水肺潜水活动，不包括在基本度假费用以内。

exertion n. /ɪg'zɜːʃn/ [U] (also exertions [pl.]) physical or mental effort; the act of making an effort 努力；尽力；费力

She was hot and breathless from the **exertion** of cycling uphill. 她骑车上山累得全身发热，喘不过气来。

conventional a. [usually before noun] following what is traditional or the way sth has been done for a long time 传统的；习惯的

conventional methods/approaches 传统方法

➤ 今日句子分析

主句

Researchers at the University of California, Los Angeles,

主语 定语

got round this by augmenting

谓语 宾语 方式状语

a cable-resistance training machine with a VR headset

augment 的宾语 方式状语

定语从句

running a tower-defence game (in which the player

定语 从句状语 从句主语

must protect a stronghold from waves of virtual enemies),

谓语 宾语 状语

so that pull downs, deadlifts and other moves triggered

主语

谓语

different attacks.

宾语

intensity between different games is greater than that between real and virtual exercise. The 2024 study, for example, found that “Beat Saber” counted as only light exercise (2-3 METs). And even vigorous games such as “The Thrill of the Fight” involve only arm movements and squats. Virtual objects cannot put up any real-world resistance, so VR games are no good for strength training.

但并非所有VR游戏的效果都相同。不同游戏之间的运动强度差异，甚至比真实运动与虚拟运动之间的差异还要大。例如，2024年的一项研究发现，“Beat Saber”仅属于轻度运动（2-3 METs）。即使是“The Thrill of the Fight”这类高强度游戏，也仅涉及手臂动作和深蹲。虚拟物体无法施加任何现实世界的阻力，因此VR游戏并不适合力量训练。

【 Para.6 】

Researchers at the University of California, Los Angeles, **got round** this by **augmenting** a cable-resistance training machine with a VR headset running a tower-defence game (in which the player must protect a **stronghold** from waves of virtual enemies), so that pull-downs, **deadlifts** and other moves triggered different attacks. Two groups trained for four weeks, one with the VR **add-on**, the other without. Members of the VR group showed greater improvement in strength, endurance and peak leg power. And they perceived their **exertion** level to be lower than it actually was.

加州大学洛杉矶分校的研究人员则提供了一种创新方案，解决了这一问题：他们在缆绳阻力训练机上加装了运行塔防游戏的VR头显（游戏中，玩家必须看守据

句子分析：

主句：

主语 -Researchers；定语 -at the University of California, Los Angeles；谓语 -got round（这个词组及物，get round sth “解决某个问题”，且一般是通过“较为迂回的方式”解决/绕开问题本身）；宾语 -this；方式状语 -by augmenting a cable-resistance training machine；

方式状语 -with a VR headset；定语，修饰VR headset-running a tower-defence game；

定语从句，修饰game：

状语 -in which（=in the game）；主语 -the player；谓语 -must protect；宾语 -a stronghold；状语 -from waves of virtual enemies (protect sb from sth “保护……，不受……的伤害”)；

结果状语从句：

（连词）-so that；从句主语 -pull-downs, deadlifts and other moves；谓语 -triggered；宾语 -different attacks.

➤ 写作积累

Para.1 “The Thrill of the Fight” is a virtual boxing game that works up a real sweat.

work up a sweat属于固定搭配，意为“累得满头大汗”，与break a sweat/burn calories相近，但前者更强调“费力气”，后者更偏向“减肥”。

其中work up属于多义词组，除了这个搭配外，还可以搭配：

1. work up courage/enthusiasm等 鼓起勇气/培养出热情，“激发……（某种情感或者状态）”，例：

I can't work up any enthusiasm for his idea. 我对他的想法怎么也热心不起来。

2. work oneself up “令……发怒”，例：

She worked herself up into a bit of a state. 她把自己搞得有点动肝火。

Para.5 Virtual objects cannot put up any real-world resistance, so VR games are no good for strength training.

put up+ resistance，这里的put up是“进行/开展”的意思，因此这个词组是指“进行抵抗”的意思，在这个语境下，则是指“有抗阻训练的效果”，而put up本身也有多重含义，以下列举几个常见的用法：

1. put up+ money “提供资金”，The state agreed to put up \$69,000 to start his company. 国家同意出资\$69000创办他的公司。

2. put sb up “收留某人”，I wanted to know if she could put me up for a few days. 我想知道她能否让我留宿几天。

3. put up with sth/sb “忍受……”，They had put up with behaviour from their son which they would not have tolerated from anyone else. 他们容忍了自己儿子的行为，而换作别的任何人，他们才不会容忍。

点，抵御一波波虚拟敌人的进攻），使下拉、硬拉等动作能偶触发不同的攻击。两组受试者进行了为期四周的训练，一组使用VR辅助设备，另一组则没有。VR组成员在力量、耐力和腿部峰值功率方面的提升更为显著。而且，他们主观上感觉的运动强度低于实际强度。

【 Para.7 】

Several other projects have found the same thing. People exercising in VR consistently underestimate how hard they are working. They also find it more fun than **conventional** exercise. But does this make them more likely to keep doing it? The evidence is

unclear. Beyond this research, VR-exercise studies are often small and do not use comparisons or controls that allow their effects to be assessed. All that said, you don't need a trial to show that VR exercise is better than none.

其他几项研究也得出了相同的结论。在虚拟现实锻炼的人总是会低估自己的运动强度。他们还觉得这比传统锻炼更有趣。但这会让他们更愿意坚持下去吗？证据尚不明确。除了这项研究之外，关于虚拟现实锻炼的研究通常规模较小，且未采用能够评估其效果的对照组或对照实验。尽管如此，无需试验也能证明，进行虚拟现实锻炼总比完全不锻炼要好。